

- 1) Find the gradients of the tangents to the following curves at the given points. $y = x^2 + 2x + 1$ (2, 9)

2) Differentiate: $y = \left(3x - \frac{1}{x}\right)^2$

$$y = \frac{1}{x} - \frac{1}{x^2} + \pi$$

- 3) The number of people, N , at a shopping centre t hours after 08:00 is modelled by
 $N = 20t^2 + 80t + 1000$ for $0 \leq t \leq 4$

a Find ____

$$\frac{dN}{dt}$$

b Calculate the rate at which people are arriving at the shopping centre in people/hr at

i 08:00 ii 10:00 iii 11:45

4) Solve for shape of the Curve. and Equation of tangent

$$y = 8 - 12x - 2x^2, \quad x = 2$$